Differential Equations

ANSWER KEY

Section 1: Separation of Variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$.

a. $y^2 = 2x^2 + 16$ ✓

b. $y^2 = x^2 + 16$

c. $y = 2x + 4$

d. $y^2 = 2x^2 + 4$

Separate variables: $y\,dy = 2x\,dx$. Integrate: $\frac{y^2}{2} = x^2 + C \Rightarrow y^2 = 2x^2 + 2C$. At $(0, 4)$: $16 = 0 + 2C \Rightarrow 2C = 16$. Solution: $y^2 = 2x^2 + 16$.

2 Find the general solution of $\frac{dy}{dx} = xy$.

a. $y = Ae^{x^2/2}$ ✓

b. $y = Ae^{x^2}$

c. $y = \frac{1}{2}x^2 + C$

d. $y = Ae^{x}$

$\frac{1}{y}dy = x\,dx \Rightarrow \ln|y| = \frac{x^2}{2} + c \Rightarrow y = Ae^{x^2/2}$.

3 Solve the differential equation $\frac{dy}{dx} = 1 + y^2$.

a. $y = \tan(x + C)$ ✓

b. $y = \ln(1 + x) + C$

c. $y = e^{x} - 1$

d. $y = \arctan(x) + C$

$\frac{1}{1+y^2}dy = dx \Rightarrow \arctan(y) = x + C \Rightarrow y = \tan(x + C)$.

Section 2: First-Order Equations Requiring Partial Fractions

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4 Find the general solution of $\frac{dy}{dx} = y(1-y)$.

a. $y = \frac{Ae^x}{1+Ae^x}$ ✓

b. $y = Ae^x - A$

c. $y = \ln|1-y| + x$

d. $y = \frac{1}{1+Ae^{-x}}$

$\frac{1}{y(1-y)}dy = dx \Rightarrow \left(\frac{1}{y} + \frac{1}{1-y}\right)dy = dx \Rightarrow \ln|y| - \ln|1-y| = x + c \Rightarrow \ln\left|\frac{y}{1-y}\right| = x + c \Rightarrow \frac{y}{1-y} = Ae^x \Rightarrow y = \frac{Ae^x}{1+Ae^x} = \frac{1}{1+A^{-}}$

5 Solve $\frac{dy}{dt} = k(M-y)$ with $y(0) = 0$ for constant $k, M > 0$.

a. $y = M(1-e^{-kt})$ ✓

b. $y = Me^{-kt}$

c. $y = M - e^{kt}$

d. $y = M(1+e^{-kt})$

$\frac{1}{M-y}dy = kdt \Rightarrow -\ln|M-y| = kt + c \Rightarrow M - y = Ae^{-kt}$. Since $y(0) = 0$, $A = M$. Thus $y = M(1-e^{-kt})$.

6 A chemical reaction produces a substance according to $\frac{dx}{dt} = 2(4-x)^2$ with $x(0) = 0$. Find $x(t)$.

a. $x(t) = 4 - \frac{4}{8t+1}$ ✓

b. $x(t) = 4(1-e^{-2t})$

c. $x(t) = \frac{8t}{2t+1}$

d. $x(t) = 4 - \frac{1}{2t}$

$\frac{1}{(4-x)^2}dx = 2dt \Rightarrow \frac{1}{4-x} = 2t + C$. At $t = 0$: $1/4 = C$. $\frac{1}{4-x} = 2t + \frac{1}{4} = \frac{8t+1}{4} \Rightarrow 4 - x = \frac{4}{8t+1} \Rightarrow x = 4 - \frac{4}{8t+1}$.

Section 3: Rate of Change Modeling Problems

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7 The rate of decay of a substance is proportional to its remaining mass m. Write down the differential equation governing $m(t)$ for constant $k > 0$.

a. $\frac{dm}{dt} = -km$ ✓

b. $\frac{dm}{dt} = km$

c. $\frac{dm}{dt} = -k$

d. $\frac{d^2m}{dt^2} = -km$

Rate of decay means rate of change is negative and proportional to mass: $\frac{dm}{dt} = -km$.

8 Water leaks from a cylindrical tank of cross-sectional area A through a hole at the bottom according to Torricelli's Law: $\frac{dh}{dt} = -c\sqrt{h}$. If initial height is H, how long does it take to empty completely?

a. $T = \frac{2\sqrt{H}}{c}$ ✓

b. $T = \frac{\sqrt{H}}{c}$

c. $T = \frac{H}{c}$

d. $T = \frac{H^2}{2c}$

$h^{-1/2}dh = -cdt \Rightarrow 2\sqrt{h} = -ct + C$. At $t = 0, C = 2\sqrt{H}$. Empty when $h = 0 \Rightarrow 0 = -ct + 2\sqrt{H} \Rightarrow t = \frac{2\sqrt{H}}{c}$.

9 A population P grows according to the logistic model $\frac{dP}{dt} = 0.04P\left(1 - \frac{P}{2000}\right)$. What is the carrying capacity of the population?

a. **2000** ✓

b. 0.04

c. 1000

d. 50000

The logistic equation is $\frac{dP}{dt} = rP(1 - P/K)$, where K is the carrying capacity. Here $K = 2000$.

Section 4: Particular Solutions & Long-Term Behavior

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10 For the logistic model in the previous question, at what population size P is the growth rate $\dfrac{dP}{dt}$ maximized?

a. $P = 1000$ ✓

b. $P = 2000$

c. $P = 500$

d. $P = 1500$

The growth rate is a quadratic function of P with zeros at 0 and K. Its vertex (maximum) occurs at exactly half the carrying capacity: $P = K/2 = 2000/2 = 1000$.

11 Solve the differential equation $x\dfrac{dy}{dx} = y + 1$ with $y(1) = 3$.

a. $y = 4x - 1$ ✓

b. $y = 3x$

c. $y = 4x^2 - 1$

d. $y = 2x + 1$

$\frac{1}{y+1}dy = \frac{1}{x}dx \Rightarrow \ln|y+1| = \ln|x| + c \Rightarrow y + 1 = Ax$. At $(1, 3)$: $3 + 1 = A(1) \Rightarrow A = 4$. Solution is $y = 4x - 1$.

12 A body cools according to $\dfrac{dT}{dt} = -0.1(T - 20)$. What does $T(t)$ approach as $t \to \infty$?

a. $20\,^\circ\text{C}$ ✓

b. $0\,^\circ\text{C}$

c. ∞

d. $-20\,^\circ\text{C}$

The solution is $T(t) = 20 + Ce^{-0.1t}$. As $t \to \infty$, $e^{-0.1t} \to 0$, so $T(t) \to 20\,^\circ\text{C}$ (ambient temperature).

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